INDUS UNIVERSITY

Mid Semester Examination

Institute	Indus Institute of Technology & Engineering	**Exam Month / Year**	March-April 2026
Program	B. Tech.	**Branch**	Mechanical
Semester	4th	**Regular / Repeater**	Regular
Course Code	ME0424	**Course Name**	Elements of Robotics (OE4)
Date	March 23, 2026	**Time**	1:00 PM to 2:30 PM
Day	Monday	**Maximum Marks**	40

Instructions:

Attempt all questions.
Make suitable assumptions wherever necessary.
Figures to the right indicate full marks.

Q 1 **Define the following terms:** **04**

1. **Robotics**
2. **Work Volume**
3. **Compliance**
4. **Precision**

Enlist the at least eight application of robots in 21st century. **04**

OR

Explain the robot anatomy with neat sketch depicting constructional features and denote names of each component. **04**

Q 2 **Draw the robot actuation system block diagram and explain in brief each block.** **04**

Show contrast between hydraulic actuation and pneumatic actuation for at least three valid points. **06**

OR

Show various modes by which a stepper motor can be operated accompanied with diagrams. **06**

State selection criteria while selecting type/mode of actuation in given application of robot. **06**

OR

Explain working of an AC servo motor with neat sketch. **06**

Q 3 **Classify the different sensors used in robotic system.** **04**

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Draw a diagram showing sensory system of a robot and explain function of each block. 06

OR

Compare human senses with the robotic sensors and state a name of a sensor used in robot which does the similar function. 06

Compare the external sensor and internal sensor of robot for at least three points. Explain how type of sensors make robotic system closed loop system. 06

OR

Discuss how sensory systems make robot alive. 06

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